

Assignment: Probability and Statistics for B.Tech ECE

20 Industrial and Electronics Data-Based Questions (2 Marks Each)

Instruction: Use proper formulas, assumptions, test statistics, and engineering interpretation.

- Q1. Bayes theorem – Fault diagnosis:** In a PCB manufacturing unit, 3% boards are faulty. An automated optical inspection system detects faults with 96% accuracy and gives 4% false alarms. For 1000 boards, use Bayes theorem to find the probability that a flagged PCB is actually faulty.
- Q2. Conditional probability – Sensor network:** In an IoT monitoring system, 65 out of 100 sensors detect temperature rise, 50 detect vibration anomaly, and 30 detect both. Find $P(T | V)$, $P(V | T)$ and comment on independence.
- Q3. Binomial distribution – Chip defects:** A VLSI fabrication line produces ICs with 8% probability of pin defect. In a lot of 15 ICs, find the probability that exactly 3 are defective and at most 2 are defective.
- Q4. Poisson distribution – Packet arrivals:** A router receives data packets at an average rate of 6 packets/ms. Using observed traffic over 20 ms, estimate the arrival model and find the probability of exactly 4 packets in the next ms.
- Q5. Normal distribution – Noise voltage:** Thermal noise voltage samples (mV) from an amplifier are: 4.8, 5.1, 4.9, 5.3, 5.0, 4.7, 5.2, 5.1, 4.9, 5.0. Assume normality and find the probability that noise exceeds 5.2 mV.
- Q6. Exponential distribution – Component lifetime:** Lifetimes (hours) of LED driver circuits are: 820, 910, 760, 880, 940, 790, 860, 900. Estimate the failure rate and probability that a new driver survives beyond 1000 hours.
- Q7. Gamma distribution – Service repair time:** Repair times (minutes) for communication modules are: 12, 15, 18, 14, 20, 17, 16, 19. Estimate gamma parameters using sample mean and variance.
- Q8. Erlang distribution – Call processing:** In a telecom switch, the waiting times until the 4th successful packet transmission are: 1.8, 2.1, 2.4, 2.0, 2.3, 1.9 ms. Fit an Erlang model and interpret the shape parameter.
- Q9. Discrete joint probability – ADC output:** Two ADC channels record quantized outputs: (1, 2), (1, 3), (2, 2), (2, 3), (3, 2), (3, 3), (2, 2), (1, 2). Construct the joint PMF, marginal PMFs, covariance, and correlation.
- Q10. Continuous joint probability – RF measurements:** Normalized amplitude-phase pairs are: (0.2, 0.3), (0.4, 0.2), (0.5, 0.1), (0.6, 0.2), (0.3, 0.4). Estimate empirically the probability that amplitude + phase < 0.7.
- Q11. CLT – Average signal strength:** Mean received powers (dBm) from repeated 64-sample windows are: -62, -61, -63, -62, -60, -61, -62, -63. Use CLT to estimate the probability that the next sample mean lies between -62.5 and -60.5.

- Q12. Hypothesis testing – Resistor production:** A resistor manufacturing plant claims mean resistance is $100\ \Omega$. Sample data: 98, 101, 100, 102, 99, 100, 101, 98, 99, 102. Formulate null and alternative hypotheses and test the claim.
- Q13. z-test – Communication latency:** For 49 measured latencies, $\bar{x} = 22$ ms, known $\sigma = 4$ ms. Test whether the true latency mean is 20 ms at 5% significance.
- Q14. t-test – Battery backup:** Battery backup times (hours) for 10 IoT nodes: 18, 19, 20, 17, 21, 18, 19, 20, 18, 22. Test whether the mean backup is 20 hours.
- Q15. Chi-square test – Bit errors:** In 600 transmitted bits, observed errors across 6 channels are: 88, 95, 110, 102, 99, 106. Test whether error occurrence is uniformly distributed.
- Q16. F-test – Process variability:** Gain values from two amplifier production lines are: A: 12.1, 12.5, 12.3, 12.6, 12.4 B: 11.8, 12.0, 11.9, 12.2, 12.1. Test equality of variances.
- Q17. Sign test – Firmware update:** Throughput improvements (Mbps) after firmware update: 2, -1, 3, 0, -2, 4, -1, 2, 3. Use sign test to determine whether median improvement is positive.
- Q18. Wilcoxon signed-rank – DSP filter:** Noise reductions (dB) before-after filter update: 5, -2, 3, -1, 4, -3, 2, -2, 1, 3. Apply Wilcoxon signed-rank test.
- Q19. Correlation – Power vs temperature:** For a transmitter, X (temperature $^{\circ}\text{C}$): 30, 35, 40, 45, 50 Y (power W): 18, 17, 16, 15, 14 Find Karl Pearson correlation and interpret physically.
- Q20. Regression – Signal attenuation:** Distance (km): 1, 2, 3, 4, 5 Received power (dBm): -40, -45, -49, -55, -60. Find both regression lines and estimate power at 6 km.